

Homework 2 - Distributed Remote A2A System

Repository

github.com/Sudashiii/dibse_industrial_computing

One-page implementation and verification summary

Implementation

- Registry with capability-based lookup
- Search, Evidence and Synthesis agents
- A2A JSON-RPC message/send over HTTP
- Docker Compose with health-gated startup

Verification

- 3 registered agents
- 2 search hits -> 2 evidence items
- 2 synthesis findings returned
- Compose config, image build and workflow passed

Relevant execution output

A2A workflow - focused execution evidence

```
question=Welche Vorteile und Grenzen haben Retrieval-Systeme?
[2026-09-15T18:27:27+00:00] COMMAND docker compose config --quiet
[2026-09-15T18:27:27+00:00] COMMAND docker compose up --build --detach --wait registry search-agent evidence-agent ...
[2026-09-15T18:27:54+00:00] COMMAND docker compose --profile workflow run --rm orchestrator --question Welche Vorte...
INFO REGISTRY_LOOKUP capability=research.search
INFO AGENT_SELECTED capability=research.search agent_id=search-agent url=http://search-agent:8101/a2a
INFO A2A_CALL step=search agent_id=search-agent
INFO REGISTRY_LOOKUP capability=research.extract_evidence
INFO AGENT_SELECTED capability=research.extract_evidence agent_id=evidence-agent url=http://evidence-agent:8102/a2a
INFO A2A_CALL step=evidence agent_id=evidence-agent
INFO REGISTRY_LOOKUP capability=research.synthesize
INFO AGENT_SELECTED capability=research.synthesize agent_id=synthesis-agent url=http://synthesis-agent:8103/a2a
INFO A2A_CALL step=synthesis agent_id=synthesis-agent
INFO WORKFLOW_COMPLETED question='Welche Vorteile und Grenzen haben Retrieval-Systeme?' agents=search-agent,evidenc...
SUMMARY agents=3 search_hits=2 evidence_items=2 synthesis_findings=2
[2026-09-15T18:27:59+00:00] COMMAND docker compose down --remove-orphans
CLEANUP compose_down=ok
```

Only registry lookups, agent calls and workflow totals are shown.

Evidence files

- execution.log: focused command and agent trace
- workflow-output.json: complete structured result
- terminal-screenshot.png: relevant output only
- README and Compose commands are reproducible

Integration flow

Each step is selected from the Registry by capability.



Registry: port 8000 | Orchestrator: one-shot Compose profile